COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	2026-08-27	**DATE SUBMITTED:**	2026-08-27
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ACTIVITY NO. 4.1 Ansible Basics

1. PROCEDURE OUTPUT

Task 1: Run elevated ad hoc commands

ansible all -m apt -a update_cache=true

What is the result of the command? Is it successful?

```
localserver@workstation:~$ ansible all -m apt -a update_cache=true
[ERROR]: Task failed: Failed to connect to the host via ssh: Warning: Pe
rmanently added '192.168.56.101' (ED25519) to the list of known hosts.
workstation@192.168.56.101: Permission denied (publickey,password).
Origin: <adhoc 'apt' task>

{'action': 'apt', 'args': {'update_cache': 'true'}, 'timeout': 0, 'async
_val': 0, 'poll': 15}

192.168.56.101 | UNREACHABLE! => {
    "changed": false,
    "msg": "Task failed: Failed to connect to the host via ssh: Warning:
 Permanently added '192.168.56.101' (ED25519) to the list of known hosts
.\r\nworkstation@192.168.56.101: Permission denied (publickey,password).
",
    "unreachable": true
}
```

- Running this command was not successful

```
localserver@workstation:~$ ansible all -m apt -a update_cache=true --become
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/python3.14', but future ins
tallation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansib
le.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
192.168.56.102 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.14"
    },
    "cache_update_time": 1787849655,
    "cache_updated": false,
    "changed": false
}
192.168.56.101 | SUCCESS => {
    "cache_update_time": 1787848488,
    "cache_updated": false,
    "changed": false
}
[WARNING]: Host '192.168.56.103' is using the discovered Python interpreter at '/usr/bin/python3.14', but future ins
tallation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansib
le.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
192.168.56.103 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.14"
    },
    "cache_update_time": 1787849945,
    "cache_updated": false,
    "changed": false
}
```

- Running the modified command is successful

Installing vim

```
localserver@workstation:~$ ansible all -m shell -a "which vim"
[WARNING]: Host '192.168.56.103' is using the discovered Python inte
tallation of another Python interpreter could cause a different inte
le.com/ansible-core/2.20/reference_appendices/interpreter_discovery.
192.168.56.103 | CHANGED | rc=0 >>
/usr/bin/vim
192.168.56.101 | CHANGED | rc=0 >>
/usr/bin/vim
[WARNING]: Host '192.168.56.102' is using the discovered Python inte
tallation of another Python interpreter could cause a different inte
le.com/ansible-core/2.20/reference_appendices/interpreter_discovery.
192.168.56.102 | CHANGED | rc=0 >>
/usr/bin/vim
```

```
192.168.56.102 | CHANGED | rc=0 >>
Sorting...
Full Text Search...
vim-nox/resolute-updates,resolute-security 2:9.1.2141-1ubuntu4.8 amd64
  Vi IMproved - enhanced vi editor - with scripting languages support

vim-tiny/resolute-updates,resolute-security,now 2:9.1.2141-1ubuntu4.8 amd64 [installed,automatic]
  Vi IMproved - enhanced vi editor - compact version
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.
[WARNING]: Host '192.168.56.103' is using the discovered Python interpreter at '/usr/bin/python3.14', but f
tallation of another Python interpreter could cause a different interpreter to be discovered. See https://d
le.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
192.168.56.103 | CHANGED | rc=0 >>
Sorting...
Full Text Search...
vim-nox/resolute-updates,resolute-security 2:9.1.2141-1ubuntu4.8 amd64
  Vi IMproved - enhanced vi editor - with scripting languages support

vim-tiny/resolute-updates,resolute-security,now 2:9.1.2141-1ubuntu4.8 amd64 [installed,automatic]
  Vi IMproved - enhanced vi editor - compact version
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.
192.168.56.101 | CHANGED | rc=0 >>
Sorting...
Full Text Search...
vim-nox/resolute-updates,resolute-security 2:9.1.2141-1ubuntu4.8 amd64
  Vi IMproved - enhanced vi editor - with scripting languages support

vim-tiny/resolute-updates,resolute-security,now 2:9.1.2141-1ubuntu4.8 amd64 [installed,automatic]
  Vi IMproved - enhanced vi editor - compact version
```

Can you describe the result of this command? Is it a success? Did it change anything in the remote servers?

- Yes, the vim was successfully installed in the remote server

Installing snapd

```
localserver@workstation:~$ ansible all -m apt -a "name=snapd state=latest" --become --ask-become-pass
BECOME password:
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/python3.14', but
tallation of another Python interpreter could cause a different interpreter to be discovered. See https://
le.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
192.168.56.102 | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.14"
    },
    "cache_update_time": 1787849655,
    "cache_updated": false,
    "changed": true,
    "stderr": "",
    "stderr_lines": [],
    "stdout": "Reading package lists...\nBuilding dependency tree...\nReading state information...\nSolvin
```

- By doing this, all of the servers will able to force update the snapd package in the remote servers.

Task 2: Writing our First Playbook

```
localserver@workstation: ~/CPE212_Duenas
 GNU nano 8.7.1
---
- hosts: all
  become: true
  tasks:

  - name: install apache2 package
    apt:
      name: apache2
```

Running the .yml file

```
localserver@workstation:~/CPE212_Duenas$ ansible-playbook --ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.101]
[WARNING]: Host '192.168.56.102' is using the discovered Python interpreter at '/usr/bin/python3.14', but future ins
tallation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansib
le.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.102]
[WARNING]: Host '192.168.56.103' is using the discovered Python interpreter at '/usr/bin/python3.14', but future ins
tallation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansib
le.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [192.168.56.103]

TASK [install apache2 package] *************************************************************************************
changed: [192.168.56.102]
changed: [192.168.56.103]
changed: [192.168.56.101]

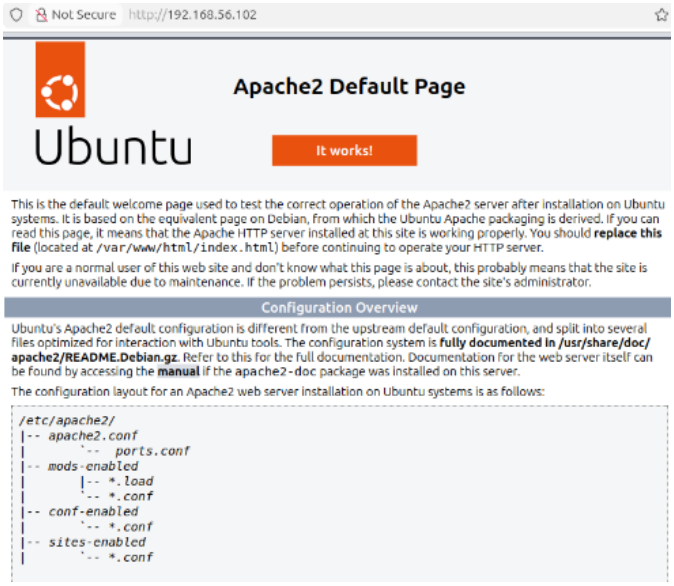
PLAY RECAP *********************************************************************************************************
192.168.56.101             : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

192.168.56.102             : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

192.168.56.103             : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0


localserver@workstation:~/CPE212_Duenas$
```

-



-

```
TASK [install apache2 package] *************************************
********
ok: [192.168.56.103]
ok: [192.168.56.102]
ok: [192.168.56.101]

TASK [install apache2 package] *************************************
********
ok: [192.168.56.103]
ok: [192.168.56.102]
ok: [192.168.56.101]

TASK [add PHP support for apache] **********************************
********
changed: [192.168.56.102]
changed: [192.168.56.101]
changed: [192.168.56.103]

PLAY RECAP *********************************************************
********
192.168.56.101             : ok=4    changed=1    unreachable=0    faile
d=0    skipped=0    rescued=0    ignored=0
192.168.56.102             : ok=4    changed=1    unreachable=0    faile
d=0    skipped=0    rescued=0    ignored=0
192.168.56.103             : ok=4    changed=1    unreachable=0    faile
d=0    skipped=0    rescued=0    ignored=0

localserver@workstation:~/CPE212_Duenas$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. What is the importance of using a playbook?
   - Playbooks matter because they help automate IT tasks. They let you run several setup and control steps in a set order. That way, you get the same setup each time. It cuts down on mistakes made by people. It also fits the idea of infrastructure as code, so you can track changes, manage versions, and scale work without losing control. Teams can also review what was done more easily.

2. Summarize what we have done on this activity.
   - For this task, I did a few basic Ansible runs first. I started with ad-hoc commands to handle packages. I updated or installed tools like vim and snapd on more than one remote machine. After that, I moved to playbooks. I wrote an install_apache.yml file. Then I ran it to set up apache2 and enable PHP support. Finally, I checked the result. I opened the Apache2 default page in the browser on the remote hosts to confirm it was working.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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